

**B.E. ELECTRICAL ENGINEERING SECOND YEAR SECOND SEMESTER
EXAM 2023**

Time: Three hours

Full Marks: 100

**POWER SUPPLY SYSTEM
PART I
Solutions**

Question 1(a)

Problem Statement: Discuss the necessity of pure water in a steam power plant and also the procedure of getting the pure water. (6+4)

Thought Process

Goal: Explain why boiler feed water must be pure and outline the demineralization process.

Methodology: 1. **Necessity:** Raw water contains dissolved salts, dissolved gases (oxygen, carbon dioxide), and suspended solids. If used directly, these cause scale formation (reducing heat transfer and causing tube bursts), corrosion (weakening metal), and carryover (silica depositing on turbine blades).

2. **Procedure:** The Demineralization (DM) process involves passing water through cation exchange resins (to remove positive ions) and anion exchange resins (to remove negative ions), followed by a mixed bed and deaeration to remove dissolved gases.

Answer:

Necessity of Pure Water:

Raw water from rivers or lakes contains suspended impurities, dissolved salts (like calcium, magnesium, and silica), and dissolved gases (like oxygen and carbon dioxide). Using raw water directly in a high-pressure boiler leads to severe operational and safety issues:

1. **Scale Formation:** Dissolved salts precipitate at high temperatures, forming a hard scale on the inner walls of boiler tubes. This scale acts as a thermal insulator, drastically reducing heat transfer efficiency and potentially causing the boiler tubes to overheat and rupture.
2. **Corrosion:** Dissolved oxygen and low pH levels lead to the oxidation and pitting of the boiler metal and feed water pipes, reducing their structural integrity and lifespan.
3. **Steam Carryover:** High concentrations of impurities (especially silica) can vaporize or be physically carried by wet steam into the superheater and turbine. Silica deposits on the high-speed turbine blades alter their aerodynamic profile, reduce turbine efficiency, and cause dangerous mechanical imbalances.

Procedure of Getting Pure Water (Demineralization):

To achieve ultra-pure feed water, thermal power plants utilize a Demineralization (DM) Plant. The standard procedure involves:

1. **Pre-treatment:** Raw water is first passed through clarifiers and sand filters to remove suspended solids and organic matter.
2. **Cation Exchange:** The filtered water passes through a strong acid cation exchange resin bed. Here, positive metallic ions (like Calcium, Magnesium, and Sodium) are absorbed by the resin and replaced with Hydrogen ions (H^+).
3. **Anion Exchange:** The acidic water then passes through a strong base anion exchange resin bed. Negative ions (like Chlorides, Sulfates, and Silicates) are absorbed and replaced with Hydroxyl ions (OH^-). The H^+ and OH^- ions combine to form pure water (H_2O).
4. **Mixed Bed Polishing:** To ensure zero leakage of ions, the water passes through a mixed bed containing both cation and anion resins, which acts as a final polishing stage.
5. **Deaeration:** Finally, the demineralized water is heated in a deaerator to drive off dissolved oxygen and carbon dioxide, completely preparing it for the boiler.

Question 1(b)

Problem Statement: Justify the statement-‘Steam power plants are not suitable for supplying fluctuating loads.’

Thought Process

Goal: Provide technical reasons why steam plants are operated as base load rather than peak/variable load.

Methodology: Focus on the thermal inertia of the system. Boilers hold hundreds of tons of water and metal. Changing load requires changing the combustion rate, boiling rate, and steam pressure, which takes hours. Furthermore, running at partial loads severely degrades thermodynamic efficiency.

Answer:

Justification:

Steam power plants (coal or nuclear fired) are inherently designed to operate as **base load plants** (running continuously at near full capacity) and are fundamentally unsuited for supplying rapidly fluctuating loads due to the following engineering constraints:

1. **High Thermal Inertia:** A steam power plant possesses massive thermal inertia. The boilers contain hundreds of tons of water and heavy steel structures. To respond to a sudden increase in electrical load, the plant must burn more coal, heat the massive volume of water, and build up high-pressure steam. This entire thermodynamic chain reaction is extremely slow, often taking several hours to safely ramp up or ramp down power output.
2. **Efficiency at Part-Loads:** Steam turbines and their complex auxiliary systems (feed pumps, forced draught fans, induced draught fans) are designed and highly optimized for maximum efficiency at their rated continuous capacity. Operating the plant at lower or fluctuating loads drastically reduces the overall thermal efficiency, increasing the fuel cost per unit of electricity generated.
3. **Thermal Stresses:** Rapidly changing the firing rate to follow fluctuating loads induces severe thermal expansion and contraction cycles (thermal fatigue) in the thick metal walls

of the boiler drum, superheater headers, and turbine casings, significantly reducing the lifespan of these extremely expensive components.

Question 2

Problem Statement: Discuss the functions of (i) Control rod of nuclear reactor (ii) starting motor of gas turbine power plant (iii) surge tank for hydroelectric turbine (iv) DM plant of steam power station (v) turning gear of steam power plant.

Thought Process

Goal: Define the precise functional role of five specific power plant components.

Answer:

1. **Control rod of a nuclear reactor:** Made of neutron-absorbing materials (such as Boron or Cadmium), control rods are inserted or withdrawn from the reactor core to regulate the rate of nuclear fission. By absorbing excess neutrons, they control the thermal power output and are completely inserted to instantly shut down (scram) the reactor during emergencies.
2. **Starting motor of a gas turbine power plant:** A gas turbine is not self-starting. The starting motor mechanically cranks the main shaft to rotate the air compressor up to a minimum required speed (usually 20% to 30% of rated speed). This ensures sufficient compressed air is supplied to the combustion chamber so that fuel ignition can occur and the turbine becomes self-sustaining.
3. **Surge tank for a hydroelectric turbine:** An open-topped tank or vertical standpipe located near the turbine between the reservoir and penstock. Its function is to absorb the destructive high-pressure waves (water hammer) caused by the sudden closing of turbine gates upon load rejection, and to provide an immediate reservoir of water when the load suddenly increases.
4. **DM (Demineralization) plant of a steam power station:** Its function is to completely remove all dissolved mineral salts, silica, and ionic impurities from raw water using ion-exchange resins. This produces ultra-pure feed water, which is essential to prevent scaling on boiler tubes and to prevent silica deposits on the turbine blades.
5. **Turning gear of a steam power plant:** An electric motor-driven mechanism engaged during the shutdown phase of the plant. Its function is to continuously rotate the massive steam turbine shaft at a very slow speed (usually 3 to 30 RPM) while it cools down. This prevents the heavy rotor from warping or bowing due to uneven thermal contraction.

Question 3

Problem Statement: With the help of a suitable diagram, discuss the turbine governing system used in hydroelectric plant.

Thought Process

Goal: Illustrate and explain how hydro turbine speed is kept constant via governing.

Methodology: 1. **Diagram:** Draw a block diagram showing the Speed Sensor (Flyball or electronic) → Control Mechanism (Pilot Valve) → Hydraulic Actuator (Servomotor) → Water Flow Control (Wicket Gates/Spear).

2. **Operation:** Explain that when load drops, speed rises. The sensor detects this, signals the pilot valve to direct high-pressure oil into the servomotor, which physically closes the wicket gates to reduce water flow, restoring nominal speed.

Answer:

Schematic Block Diagram of a Hydroelectric Governor System:

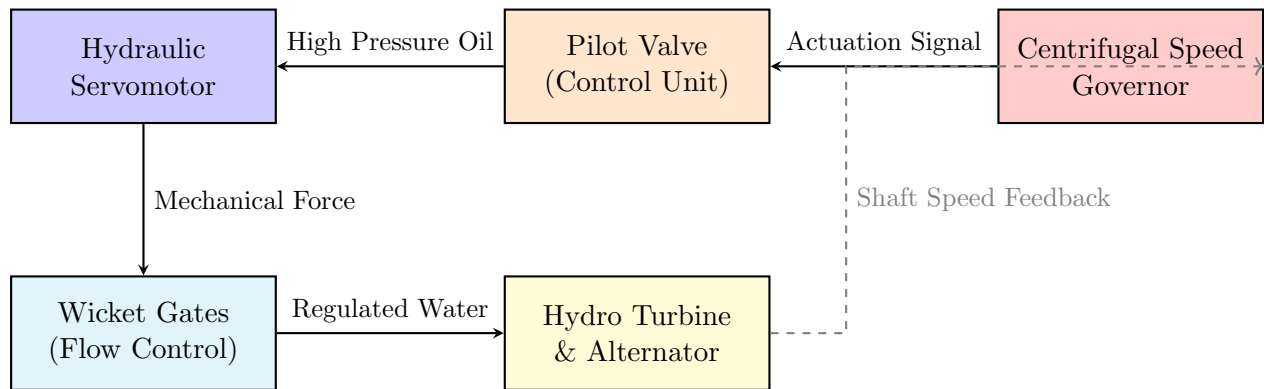


Figure 1: Block Diagram of a Hydroelectric Turbine Governing System

Discussion of the Governing System:

The primary objective of the governing system is to maintain the alternator's rotational speed (and consequently, the electrical grid frequency) absolutely constant, regardless of electrical load variations. It achieves this by automatically adjusting the quantity of water flowing into the turbine.

1. **Speed Sensing:** The centrifugal speed governor is mechanically or electrically coupled to the main turbine shaft. It continuously monitors the rotational speed.
2. **Load Rejection Scenario (Speed Increase):** If the electrical load on the grid suddenly drops, the mechanical torque exceeds the electrical counter-torque, causing the turbine shaft to accelerate. The flyballs of the centrifugal governor fly outward due to increased centrifugal force.
3. **Control Actuation:** This outward mechanical movement pulls a linkage that actuates the **Pilot Valve**. The pilot valve is a highly sensitive hydraulic switch that directs highly pressurized hydraulic oil from an accumulator into the specific chamber of the **Hydraulic Servomotor**.
4. **Flow Regulation:** The massive hydraulic pressure forces the piston of the servomotor to move. This piston is mechanically linked to the **Wicket Gates** (in reaction turbines) or the spear valve (in impulse turbines). The movement partially closes the gates, restricting the volume of water striking the turbine blades. The mechanical input power drops, matching the new electrical load, and the turbine returns to its normal synchronous speed.

Question 4

Problem Statement: A certain region has a maximum demand of 500 MW which decrease linearly with time to 200MW over a year. The supply utility has to meet the demand by setting up a gas-turbine and a coal fired thermal power plant. The installation and operating costs of the two kinds of plants are as follows:

Gas-turbine plant: Rs. 80 lakhs, Rs. 1000 per kW and Rs. 2.00 per kWh

Coal-fired plant: Rs. 100 lakhs, Rs. 6000 per kW and Rs. 1.20 per kWh

Determine the (i) annual generation of each plant and (ii) overall cost per kWh. Derive necessary formula. (10)

Thought Process

Goal: Optimize the capacity mix of Base (Coal) and Peak (Gas) plants and calculate total energy and costs.

Methodology: 1. **Derive Optimum Hours (H):** Set the annual cost of 1 kW of Gas equal to 1 kW of Coal to find the break-even operating hours. 2. **Load Duration Curve (LDC) Geometry:** The LDC is a straight line from 500 MW ($t = 0$) to 200 MW ($t = 8760$). 3. **Capacity Allocation:** Find the power level on the LDC corresponding to the optimum hours H . This sets the capacity boundary between the base and peak plants. 4. **Energy Calculation:** Area under the curve. Peak energy is a triangle. Base energy is the remaining trapezium/rectangle. 5. **Cost Calculation:** Apply the given rates to the calculated capacities and energy totals.

Answer:

Derivation of Optimum Operating Time (H):

Let the total annual cost C of a power plant of capacity P (in kW) operating for t hours per year generating energy E (in kWh) be represented by:

$$C = C_{fixed} + A \cdot P + B \cdot E \quad (1)$$

Where C_{fixed} is the fixed standing cost, A is the capital cost per kW, and B is the running cost per kWh. Since $E = P \cdot t$, for a 1 kW unit ($P = 1$), the variable annual cost depends strictly on t : Cost per kW = $A + B \cdot t$.

To find the most economical division of load, we find the break-even operating hours (H) where the cost of generating 1 kW from the Peak Plant (Gas) equals the cost from the Base Plant (Coal).

$$\begin{aligned} A_{gas} + B_{gas} \cdot H &= A_{coal} + B_{coal} \cdot H \\ H(B_{gas} - B_{coal}) &= A_{coal} - A_{gas} \\ H &= \frac{A_{coal} - A_{gas}}{B_{gas} - B_{coal}} \end{aligned}$$

Step 1: Calculate Optimum Operating Hours (H)

Given parameters:

- Gas Plant (Peak): $A_{gas} = 1000$ Rs/kW, $B_{gas} = 2.00$ Rs/kWh
- Coal Plant (Base): $A_{coal} = 6000$ Rs/kW, $B_{coal} = 1.20$ Rs/kWh

$$H = \frac{6000 - 1000}{2.00 - 1.20} = \frac{5000}{0.8} = \mathbf{6250 \text{ hours}}$$

Any load required for more than 6250 hours per year should be supplied by the base load (coal) plant.

Step 2: Determine Plant Capacities from the Load Duration Curve

The Load Duration Curve (LDC) decreases linearly from $P_{max} = 500$ MW to $P_{min} = 200$ MW over a total time $T = 8760$ hours. The equation for the load $P(t)$ in MW at any time t is:

$$P(t) = 500 - \left(\frac{500 - 200}{8760} \right) \cdot t = 500 - \frac{300}{8760} \cdot t$$

The capacity boundary P_{base} occurs at $t = 6250$ hours:

$$P_{base} = 500 - \left(\frac{300}{8760} \right) \cdot 6250 = 500 - 214.041 = \mathbf{285.959 \text{ MW}}$$

Therefore:

- **Capacity of Base Load Plant (Coal) = 285.959 MW = 285,959 kW**
- **Capacity of Peak Load Plant (Gas) = 500 – 285.959 = 214.041 MW = 214,041 kW**

Step 3: Calculate Annual Generation (Energy) of Each Plant

(i) *Peak Load Plant Energy (E_{gas}):* This is the area of the upper triangle on the LDC above 285.959 MW, lasting for 6250 hours.

$$E_{gas} = \frac{1}{2} \cdot \text{Base} \cdot \text{Height} = \frac{1}{2} \cdot (6250 \text{ hours}) \cdot (214.041 \text{ MW}) = 668,878.125 \text{ MWh} = \mathbf{6.688 \times 10^8 kWh}$$

(ii) *Total Energy Demand (E_{total}):* Area of the entire trapezium.

$$E_{total} = \frac{1}{2} \cdot (500 + 200) \text{ MW} \cdot 8760 \text{ hours} = 3,066,000 \text{ MWh} = \mathbf{3.066 \times 10^9 kWh}$$

(iii) *Base Load Plant Energy (E_{coal}):*

$$E_{coal} = E_{total} - E_{gas} = 3,066,000 - 668,878.125 = 2,397,121.875 \text{ MWh} = \mathbf{2.397 \times 10^9 kWh}$$

Step 4: Calculate Overall Cost per kWh

Let us compute the total annual cost. Note: 1 Lakh = 100,000 Rs.

$$\begin{aligned} \text{Total Cost of Gas Plant} &= \text{Fixed} + (\text{Cap. Rate} \cdot \text{kW}) + (\text{Run. Rate} \cdot \text{kWh}) \\ &= 8,000,000 + (1000 \cdot 214,041) + (2.00 \cdot 668,878,125) \\ &= 8,000,000 + 214,041,000 + 1,337,756,250 = \mathbf{1,559,797,250 \text{ Rs.}} \end{aligned}$$

$$\begin{aligned} \text{Total Cost of Coal Plant} &= 10,000,000 + (6000 \cdot 285,959) + (1.20 \cdot 2,397,121,875) \\ &= 10,000,000 + 1,715,754,000 + 2,876,546,250 = \mathbf{4,602,300,250 \text{ Rs.}} \end{aligned}$$

$$\text{Grand Total Annual Cost} = 1,559,797,250 + 4,602,300,250 = \mathbf{6,162,097,500 \text{ Rs.}}$$

$$\text{Overall Cost per kWh} = \frac{\text{Grand Total Cost}}{\text{Total Energy}} = \frac{6,162,097,500}{3,066,000,000} = \mathbf{2.0098 \text{ Rs/kWh}}$$

Question 5

Problem Statement: With a schematic diagram describe the principle of operation of a close cycle gas turbine plant.

Thought Process

Goal: Illustrate and describe a Brayton cycle where the working fluid does not mix with combustion gases.

Methodology: Draw an orthogonal closed loop: Compressor → External Heater → Turbine → Pre-cooler → back to Compressor. Explain how heat is added externally, preventing turbine blade fouling and allowing use of cheaper fuels.

Answer:

Schematic Diagram of a Closed Cycle Gas Turbine Plant:

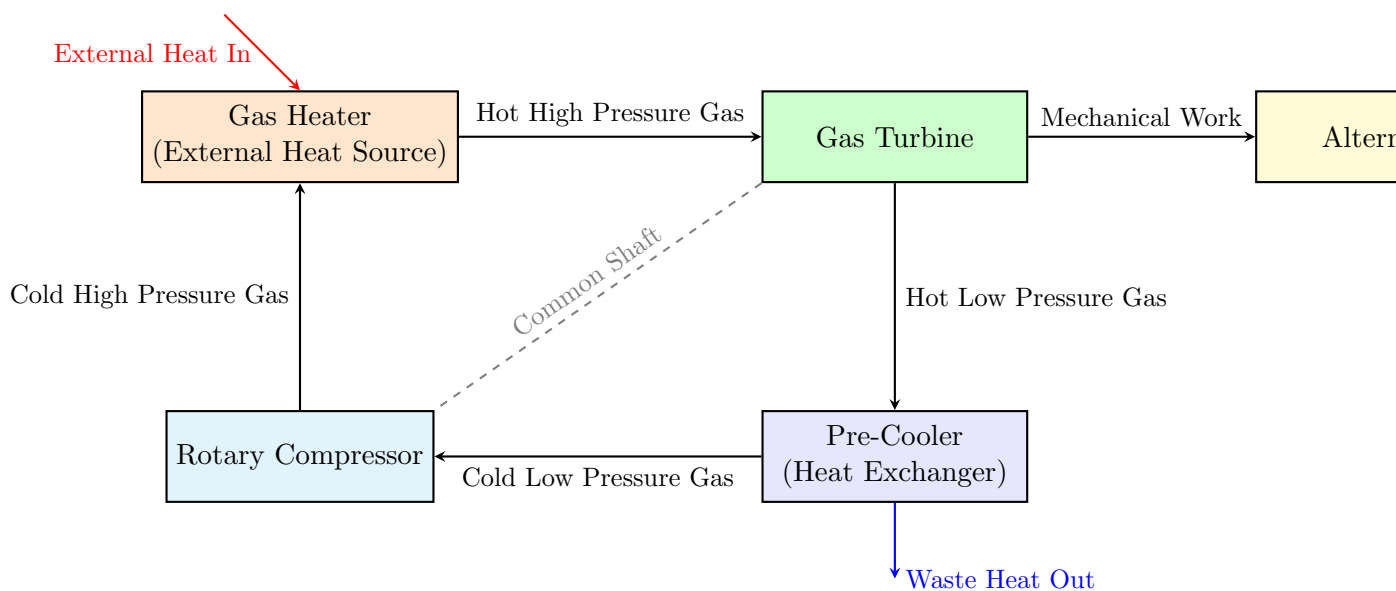


Figure 2: Closed Cycle Gas Turbine Configuration

Principle of Operation:

A closed-cycle gas turbine operates on the constant-pressure Joule/Brayton cycle. Unlike an open cycle, the working fluid (which can be pure air, Helium, or Argon) is completely sealed within the system and circulates continuously.

1. **Compression:** The working fluid enters the rotary compressor at low pressure and low temperature. The compressor performs mechanical work to compress the gas, significantly raising its pressure.
2. **External Heating:** The high-pressure gas is routed into an external gas heater (a heat exchanger). Here, heat from an external combustion furnace (or nuclear reactor) is transferred to the working fluid across tube walls. Because the fuel and combustion products never mix with the working fluid, cheap solid fuels like coal can be used without risking turbine blade erosion.
3. **Expansion:** The super-heated, high-pressure gas expands rapidly through the gas turbine. This expansion turns the turbine blades, converting thermal energy into mechanical shaft power to drive both the alternator (for electricity) and the compressor.

4. **Cooling:** The expanded, hot, low-pressure gas exits the turbine and passes through a pre-cooler. External cooling water absorbs the waste heat, reducing the temperature of the working fluid back to its initial low state before it re-enters the compressor, completing the sealed cycle.

Question 6

Problem Statement: Draw a neat diagram of boiling water nuclear reactor and describe the function of each component of such reactor.

Thought Process

Goal: Illustrate a single-loop nuclear reactor (BWR) and list component functions.

Methodology: Draw an orthogonal diagram with a reactor vessel acting as both the fission chamber and the boiler. Steam goes directly to the turbine, then to the condenser, then pumped back. Explain the core, control rods, and lack of a secondary heat exchanger.

Answer:

Diagram of a Boiling Water Reactor (BWR):

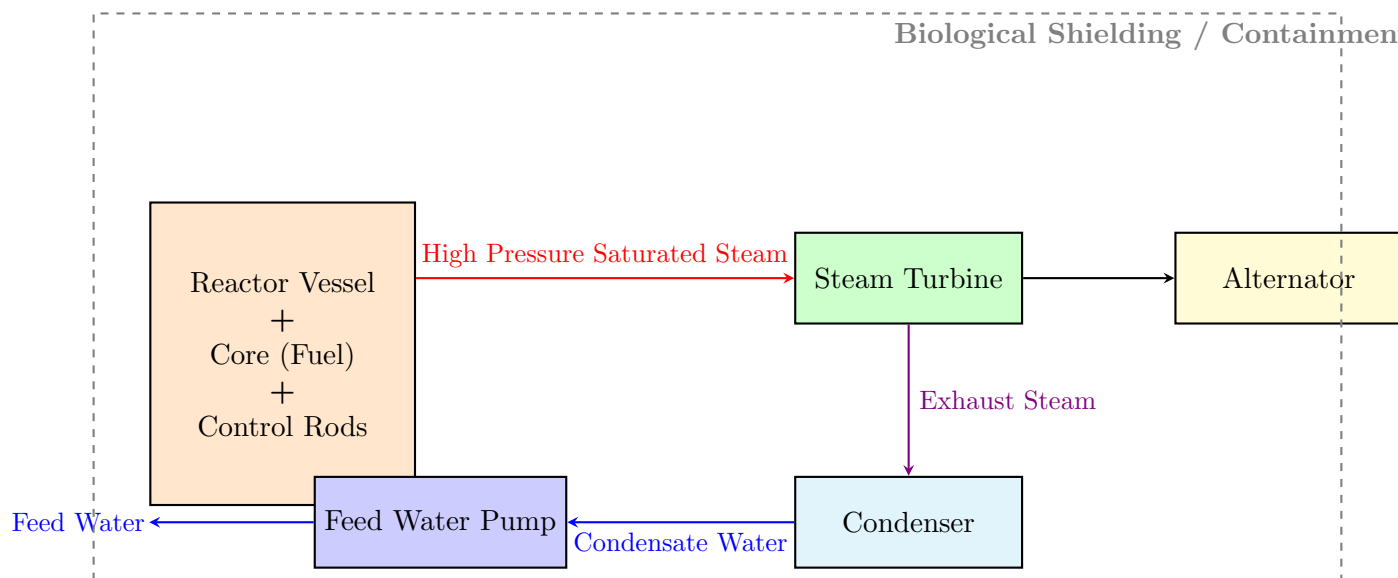


Figure 3: Schematic of a Boiling Water Reactor (BWR)

Functions of Components in a BWR:

The distinguishing feature of a Boiling Water Reactor is its direct-cycle design; the same light water acts as the coolant, the moderator, and the working fluid. There is no separate steam generator.

1. **Reactor Vessel:** A massive, high-strength steel pressure vessel that houses the nuclear core and water. It is designed to safely contain the immense pressure of the boiling water.
2. **Nuclear Core (Fuel):** Consists of fuel assemblies containing enriched Uranium-235. The nuclear fission of this fuel generates the intense thermal energy required to boil the water directly inside the vessel.

3. **Moderator / Coolant (Light Water):** The ordinary water surrounding the fuel rods slows down (moderates) the fast neutrons generated by fission so the chain reaction can be sustained. Simultaneously, it absorbs the heat and vaporizes into steam.
4. **Control Rods:** Usually inserted from the bottom of the vessel in a BWR. They contain neutron-absorbing materials. Adjusting their depth controls the reaction rate and thermal output of the reactor.
5. **Steam Turbine and Condenser:** The saturated steam generated inside the reactor vessel flows directly to the steam turbine to generate electricity. After expansion, the slightly radioactive steam is cooled back into liquid water by the condenser and pumped directly back into the reactor vessel.

Question 7

Problem Statement:

- a) Distinguish between peak load plant and base load plant. (4)
 - b) The variation of the load over the 24 hours of a day is as given below. Draw the load curve and load-duration curve: (6)
- (Table: 0-4h: 40MW, 4-8h: 60MW, 8-12h: 100MW, 12-16h: 80MW, 16-20h: 100MW, 20-24h: 70MW).

Thought Process

Goal: Distinguish plant types conceptually and accurately plot specific daily load data.
Methodology: 1. Use a comparative list or table to distinguish peak vs base based on operating hours, start-up time, and economics. 2. For the graphs, construct the step-coordinates exactly matching the table provided. Load duration requires sorting the loads in descending order to accumulate total hours.

Answer:

a) Distinction Between Base Load Plant and Peak Load Plant:

- **Base Load Plant:** These plants operate continuously to cater to the constant minimum electrical demand of the grid. They have a very high load factor. They are characterized by massive capital installation costs but very low operating/fuel costs. Because of large thermal inertia, they cannot start or stop quickly. Examples include Nuclear Power Plants and large Coal-fired Steam Power Plants.
- **Peak Load Plant:** These plants operate only during specific hours of the day when electrical demand spikes above the base level. They have a low load factor. They are characterized by lower capital costs but very high operating/fuel costs. They must have the capability for extremely rapid start-up and synchronization. Examples include Gas Turbine Plants and Pumped Storage Hydro Plants.

b) Load Curve and Load-Duration Curve:

Data Preparation for Load Duration Curve (Sorted Descending):

- 100 MW persists for 8 hours (8-12 and 16-20)
- 80 MW persists for 4 hours (12-16)

- 70 MW persists for 4 hours (20-24)
- 60 MW persists for 4 hours (4-8)
- 40 MW persists for 4 hours (0-4)

